HW 8 Part 1: Art, AI, and Storytelling

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Artist and Piece I Picked

I went with Vanessa Rosa and her piece *Little Martians: A Brief History of Consciousness in the Simulation* (2022), featured in the NVIDIA AI Art Gallery. Vanessa Rosa is a Brazilian visual artist based in the United States who blends physical media with digital tools to tell sci-fi stories. The piece is a 3D-rendered video built around her handmade ceramic heads. The story frames the heads as memorabilia from a long-lost humanity drifting through the galaxy, narrating fragments of life in our solar system.

Project page: https://www.nvidia.com/en-us/research/ai-art-gallery/artists/vanessa-rosa/
Artist site: https://www.littlemartians.world/

How She Actually Built It

Three concrete technical details from her workflow stood out to me.

1. Physical sculptures get turned into geometry through 3D scanning. She makes the ceramic heads by hand, then 3D-scans them to produce mesh assets. That output is the same kind of vertex-and-triangle data we have been working with all semester. After scanning, the meshes are imported into Blender for rigging and animation.

2. NVIDIA Audio2Face drives the facial animation from an AI-generated voice. Instead of hand-keying mouth shapes per frame, she runs an AI text-to-speech voice through Audio2Face, which uses a neural network to predict facial mesh deformations that match the audio. The output is still a deforming mesh, so the rendering side stays inside the classic pipeline, but the animation curves themselves come from an inference step rather than from an animator on a timeline.

3. A second AI pass acts on rendered video frames as a post-process. After Blender finishes rendering, she runs select frames through an AI text-to-image style transfer and uses those frames as guides to apply a stylized look across the rest of the sequence. That step is purely pixel-based and happens after rasterization is done.

How This Relates to What We Have Been Learning

The first half of her pipeline maps cleanly onto what we covered. The 3D-scanned ceramic mesh becomes vertex attributes that feed the vertex shader, primitive assembly stitches the triangles together, the rasterizer turns those triangles into fragments, and the fragment shader produces the final color. The Audio2Face step is interesting because it sits before the pipeline runs at all. It is generating the per-frame mesh deformations that will then be drawn the normal way, so the actual rendering stage is still doing what we have been doing in WebGL with uniform mat4 transforms and shaded surfaces.

The style transfer pass is where she breaks from the classic pipeline. Once the rasterizer has produced a framebuffer, that output gets handed to a neural network that re-paints it. There is no geometry being shaded at that point. It is closer to the image-processing fragment shaders we worked with in the texture lab, except the operation is not a kernel I write in GLSL; it is a learned model running outside the pipeline.

What I Took Away From This

What I keep coming back to is that her workflow is not really AI replacing graphics. It is AI plugged in at specific spots in a workflow that is otherwise very traditional. The geometry pipeline still does the heavy lifting; AI fills in the slow human bottlenecks like voicing characters, animating faces, and applying a coherent visual style across frames. The material we have been learning still describes how the image actually gets drawn, and the new tools mostly generate inputs to that process or operate on its outputs.